

Test Economics Document

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Chapter 1

Test Economics Document

Chapter 2

Introduction

This is a minimal test document to verify PDF link preprocessing works correctly.

2.1 Link Test Cases

2.1.1 Internal Cross-References (Work in PDF)

These use Quarto's `@sec-` cross-reference system. In PDF, clicking these navigates within the document:

- Link to test parameters intro: Chapter 4
- **Link to calculations appendix:** Section 5.1

Tip

Standard markdown hyperlinks only work in HTML. For PDF output, use `@sec-` cross-references instead.

2.1.2 External Cross-Site Links (Should Become Absolute URLs)

These links point to files that are NOT in test config but ARE in book config, so they should be rewritten to <https://manual.WarOnDisease.org>:

- Link to economics chapter: [Economics Analysis](#)
- Link to drug development analysis: [Drug Development Cost Analysis](#)
- Link to DFDA analysis: [DFDA Cost Benefit Analysis](#)

2.1.3 Absolute URLs (Should Remain Unchanged)

These are already absolute URLs and should not be touched:

- External site: [War on Disease](#)
- DIH site: [Decentralized Institutes of Health](#)
- GitHub: [GitHub Repository](#)

2.1.4 Anchor Links (Should Remain Unchanged)

- Jump to section: [Link Test Cases](#)
- Another anchor: [Introduction](#)

2.1.5 Cross-Reference Format Comparison

Syntax	Works in HTML	Works in PDF
<code>@sec-label</code> cross-reference	Yes	Yes
Standard markdown link	Yes	No (becomes file://)
<code>[text](https://...)</code> absolute URL	Yes	Yes (opens browser)

2.2 Conclusion

To test PDF links:

1. Run: `.venv/Scripts/python.exe scripts/generate-everything-parameters-variables-calculations-references.py`
2. Run: `python scripts/render-quarto.py test --to pdf`
3. Open the generated PDF at `_site/test/test-links.pdf`
4. Click each link and verify behavior:

Link Type	Expected Result
<code>@sec-</code> cross-references	Navigate within PDF
External URLs	Open browser to that site
Anchor links (<code>#section</code>)	Jump to section in current page
Markdown links to <code>.qmd</code> files	BROKEN - opens file URL

Chapter 3

Test Parameters Document

Chapter 4

Introduction

This is a minimal parameters document for testing cross-references.

4.1 Test Parameters

Testing if using ANY variable from `_variables.yml` triggers citation warnings for ALL embedded citations:

- ADAPTABLE trial cost per patient: \$929 (95% CI: \$929-\$1.40K)
- Test metric: 100%
- Sample value: \$1,000,000

4.2 Callout Test

Testing callout rendering in PDF:

 This is a Note Callout

This tests whether callouts render correctly in both HTML and PDF output.

 Warning Example

This is a warning callout with important information about the 3.5% threshold for social change.

 Pro Tip

The treaty redirects 1% of military spending to medical research, potentially saving millions of lives.

4.3 Simple Variable Test

One simple variable to verify the system works without heavy Monte Carlo charts:

- Test value (external): \$929 (95% CI: \$929-\$1.40K)

4.4 Calculated Variable Test

Testing if calculated variables link to their calculations in the parameters appendix:

- Treaty annual funding (calculated): \$27.2B

4.5 Citation Test

Testing if citations link to references section:

- The ADAPTABLE trial achieved a cost of \$929 (95% CI: \$929-\$1.40K) per patient ([N. C. Fund 2025](#)).
- This represents a 30x cost advantage over traditional RCTs.

4.6 Cross-Chapter Anchor Links Test

Testing if anchor links from main content to specific parameter sections in the appendix work in PDF:

- Link to ADAPTABLE cost parameter: [ADAPTABLE Trial Cost per Patient](#)
- Link to military spending parameter: [Global Military Spending](#)
- Link to treaty reduction parameter: [Treaty Reduction Percentage](#)
- Link to calculated parameter: [Treaty Annual Funding \(calculated\)](#)
- Link to appendix overview: [Parameters Overview](#)

If these work in PDF, we could potentially make variable values themselves link to their calculation sections.

4.7 Cross-Site References

These should become absolute URLs since they're NOT in the test config:

- Economics chapter: [Economics Analysis](#)
- Full analysis: [DFDA Cost Benefit Analysis](#)
- Drug development: [Drug Development Cost Analysis](#)

Chapter 5

Methodology, Parameters, and Calculations

5.1 Overview

This appendix documents all 4 parameters used in the analysis, organized by type:

- External sources (peer-reviewed): 2
- Calculated values: 1
- Core definitions: 1

5.1.1 Quick Navigation

[Calculated Values](#) (1 parameters) • [External Data Sources](#) (2 parameters) • [Core Definitions](#) (1 parameters)

5.2 Calculated Values

Parameters derived from mathematical formulas and economic models.

5.2.1 Annual Funding from 1% of Global Military Spending Redirected to DIH: \$27.2B

i Details

Annual funding from 1% of global military spending redirected to DIH

Inputs:

- [Global Military Spending in 2024](#) : \$2.72T
- [1% Reduction in Military Spending/War Costs from Treaty](#): 1%

$$\begin{aligned}
& Funding_{treaty} \\
&= Spending_{mil} \times Reduce_{treaty} \\
&= \$2.72T \times 1\% \\
&= \$27.2B
\end{aligned}$$

High confidence

5.3 External Data Sources

Parameters sourced from peer-reviewed publications, institutional databases, and authoritative reports.

5.3.1 ADAPTABLE Trial Cost per Patient: \$929

i Details

Cost per patient in ADAPTABLE trial (\$14M PCORI grant / 15,076 patients). Note: This is the direct grant cost; true cost including in-kind may be 10-40% higher.

Source: ([N. C. Fund 2025](#))

5.3.1.1 Uncertainty Range

Technical: 95% CI: [\$929, \$1.40K] • Distribution: Lognormal

What this means: There's significant uncertainty here. The true value likely falls between \$929 and \$1.40K ($\pm 25\%$). This represents a wide range that our Monte Carlo simulations account for when calculating overall uncertainty in the results.

The lognormal distribution means values can't go negative and have a longer tail toward higher values (common for costs and populations).

5.3.1.2 Input Distribution

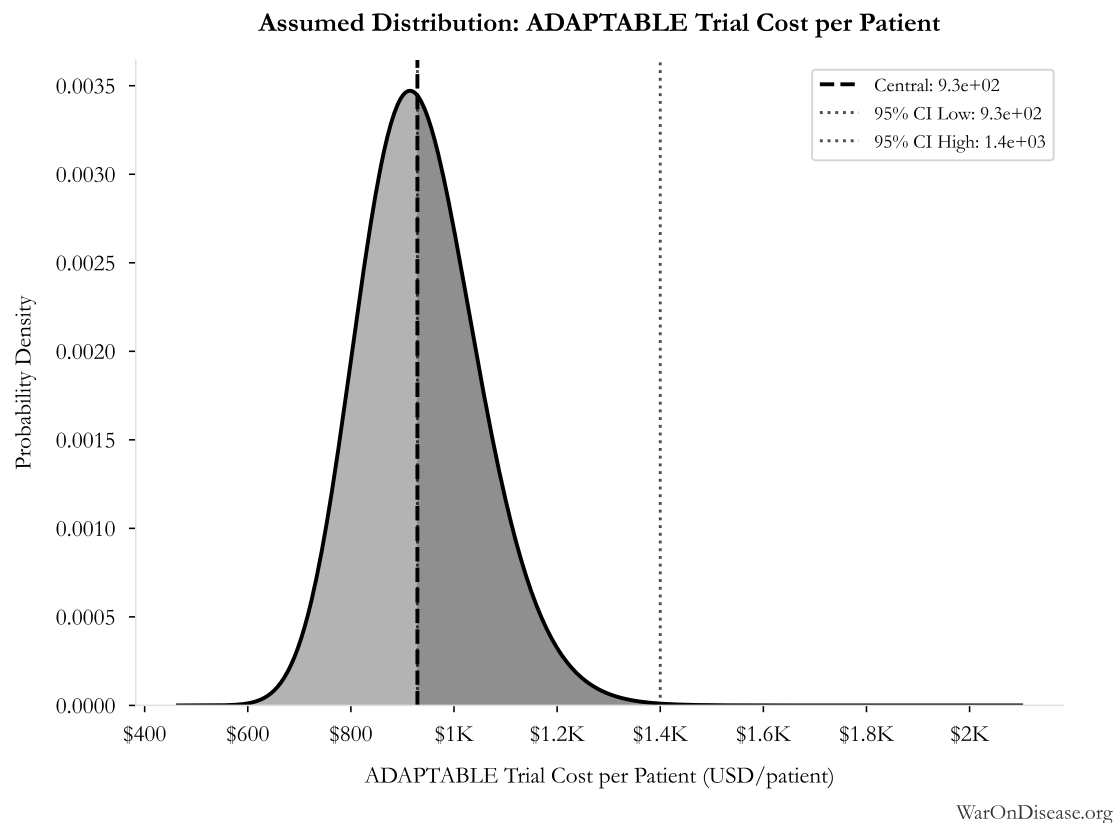


Figure 5.1: Probability Distribution: ADAPTABLE Trial Cost per Patient

This chart shows the assumed probability distribution for this parameter. The shaded region represents the 95% confidence interval where we expect the true value to fall.
~ Medium confidence

5.3.2 Global Military Spending in 2024: \$2.72T

i Details

Global military spending in 2024

Source: ([SIPRI 2025](#))

5.3.2.1 Uncertainty Range

Technical: Distribution: Fixed

High confidence

5.4 Core Definitions

Fundamental parameters and constants used throughout the analysis.

5.4.1 1% Reduction in Military Spending/War Costs from Treaty: 1%

i Details

1% reduction in military spending/war costs from treaty

5.4.1.1 Uncertainty Range

Technical: Distribution: Fixed

Core definition

Chapter 6

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